SGA - Structuration – Cours 6

Thomas Chedru, Yann Moysan

Plan du cours de Structuration 2026

- Cours 5 : 22Jan26 - Yann Moysan – Sous-jacents « classiques » et gestion
  - 8h15-11h30
- Cours 6 : 29Jan26- Yann Moysan – Sous-jacents « innovants » et gestion
  - 8h15-11h30
- Cours 7 : 05Feb26 - Yann Moysan – Hybrides et Commodities
  - 13h45-17h00
- EVAL : 10Feb26 17h15-20h15 (TBC)

Contents

BNP Paribas QIS (Quantitative Investment Strategies)

- 1. Philosophy
- 2. Index structuring challenges
- 3. Operationnal setup
- 4. Range of strategies

BNP Paribas QIS

Philosophy (1/4)

- Access to a wide range of underlyings
- Transparency
- Risk control

2006 - QIS BUSINESS STARTED

Since launching our proprietary index business (now known as Quantitative Investment Strategies (QIS)) in 2006, we have continuously strived to provide our clients with high quality, robust indices, operated within a strong governance framework. Several strategies have been live for over a decade and new indices are often new versions of existing strategies, reviewed and improved in response to regulatory changes, investor preference and the latest innovations. As an Equity Derivatives powerhouse with over 25 years experience, BNP Paribas has one of the strongest infrastructures and is recognised for its execution capacities.

EUR 35BN - QIS ASSETS UNDER MANAGEMENT

EUR 5BN - QIS ASSETS UNDER MANAGEMENT IN FUND FORMAT

Combining strong execution capabilities and expertise of an investment bank along with an asset management mindset

50 - PEOPLE DEDICATED TO QIS STRATEGIES

25 QIS Structurers, 3 Quant Researchers (QIS Lab), 10 dedicated traders, Independent Validation Research team, Publication and Coordination team, External Partnerships

1300 BNP PARIBAS INDICES, 200 EU BENCHMARKS

Philosophy (2/4)

⇨ Access to five main classes of underlyings:

- Equities
- Fixed Income (Rates, Credit, FX)
- Commodities
- Funds

⇨ Four different types of needs:

1	2	3	4
Leverage	**Yield enhancement**	**Broader investment universe**	**Convexity / protection**
Leverage your exposure to rates and credit markets using liquid solutions and boost your portfolio returns	Enhance the risk adjusted return of your portfolio with access to alternative and option based strategies	Access other markets, underlyings, instruments (like derivatives) that you do not trade yourself or where you have limited access	Access a wide range of optional products to mitigate the risk of your portfolio, via defensive or hedging indices

Philosophy (3/4)

1. A WIDE UNIVERSE OF STRATEGIES

FLEXIBLE

BNP Paribas' platform offers a wide range of QIS strategies from simple market access, to alternative or option based strategies across all asset classes. You can:

- **Chose** the strategies that fit your needs
- **Rebalance** your positions when your views change
- **Or design** your own strategy

2. EASY TO INVEST

SIMPLE

Practically, you will trade a simple swap, which will deliver the performance of the QIS strategy, providing:

- **A single product**, irrespective of the number of financial instruments being synthetically traded within the strategy
- **No operational burden**, meaning more time for added-value tasks
- You can receive a **detailed reporting** with full look-through as well as consolidated data (overall exposure, greeks…)

3. DIGITAL

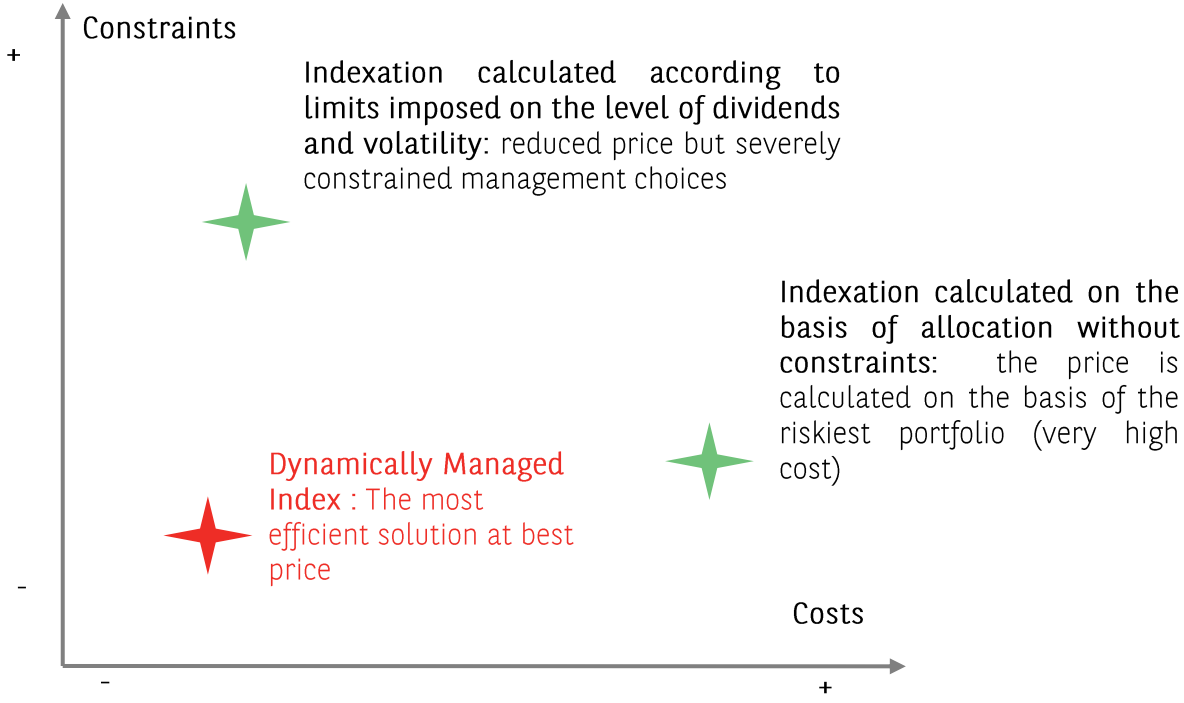
DIGITAL

BNP Paribas's digital platform (BRIO) allows you to easily:

- Have a look at BNP Paribas product suite
- **Analyze and test** the relevance of a strategy on top of your portfolio
- **Trade**, take care of **secondary orders** as well as potential **rebalancing**

Philosophy (4/4)

" You only pay for what you use"



Alternative Risk Premia Strategies (NOT EXHAUSTIVE LIST)

Overview of the range

EQUITIES	FACTOR
BNP Paribas AIR Europe Index	Volatility
BNP Paribas AIR US Index	Volatility
BNP Paribas AIR UK Index	Volatility
BNP Paribas AIR Japan Index	Volatility
BNP Paribas AIR Global Ex-Japan	Volatility
BNP Paribas TIER US Volatility Index	Volatility
BNP Paribas TIER Calendar EU Index	Volatility
BNP Paribas TIER EU 4% Index	Volatility
BNP Paribas TIER US 4% Index	Volatility
BNP Paribas TIER Switzerland 2% Index	Volatility
BNP Paribas TIER Japan Index	Volatility
BNP Paribas Volatility Mean Reversion US Index	Volatility
BNP Paribas Term Structure Volatility US Enhanced Index	Volatility
BNP Paribas Equity Mean Reversion US ER Index	Volatility
BNP Paribas Equity Mean Reversion EU/US EUR ER Index	Volatility
BNP Paribas Long Short Dividend EU ER Index	Carry
BNP Paribas Converse Spread Strategy Europe TR	Carry
BNP Paribas Converse Spread Strategy Europe USD TR	Carry
BNP Paribas Equity High Dividend Europe LS (NTR) Index	Carry
BNP Paribas Equity High Dividend US LS (NTR) Index	Carry
BNP Paribas Equity Value Europe L/S (ER) Index	Value
BNP Paribas Equity Value US L/S (ER) Index	Value
BNP Paribas Equity Momentum Europe LS (ER) Index	Momentum
BNP Paribas Equity Momentum US LS (ER) Index	Momentum
BNP Paribas Extremum reversion Risk Premia	Trend
BNP Paribas Equity Quality Europe LS (ER) Index	Defensive
BNP Paribas Equity Quality US LS (ER) Index	Defensive
BNP Paribas Equity Low Vol Europe LS (ER) Index	Defensive
BNP Paribas Equity Low Vol US LS (ER) Index	Defensive
BNP Paribas Equity GURU® Europe LS (NTR) Index	Multi-Factor
BNP Paribas Equity GURU® US LS (NTR) Index	Multi-Factor
BNP Paribas GURU® Long/Short market Neutral	Multi-Factor
BNP Paribas DEFI Equity World Market Neutral TR Index	Multi-Factor
BNP Paribas DEFI Equity World Market Neutral 2X TR Index	Multi-Factor

MULTI-ASSET	FACTOR
BNP Paribas Multi-Asset Diversified vol 8 EUR Future Index	Trend
BNP Paribas Cross Asset Trend Index	Trend
BNP Paribas Artificial Intelligence Trend 2019 Index	Trend

** Some strategies may not be available to US investors. **Ex-Agriculture & Livestock.*

COMMODITIES	FACTOR
BNP Paribas Commodity Short Variance CVXA ER Index	Volatility
BNP Paribas TIER Gold Daily Index	Volatility
BNP Paribas Commodity Short Call SCXA ER	Volatility
BNP Paribas DR Alpha ex-A&L** Index	Carry
BNP Paribas Dynamic Pre-Roll Alpha ex-A&L Target Volatility 3 Index	Carry
BNP Paribas Alpha Backwardation ex-A&L Net Index	Carry
BNP Paribas Alpha Momentum ex-A&L Net Index	Momentum

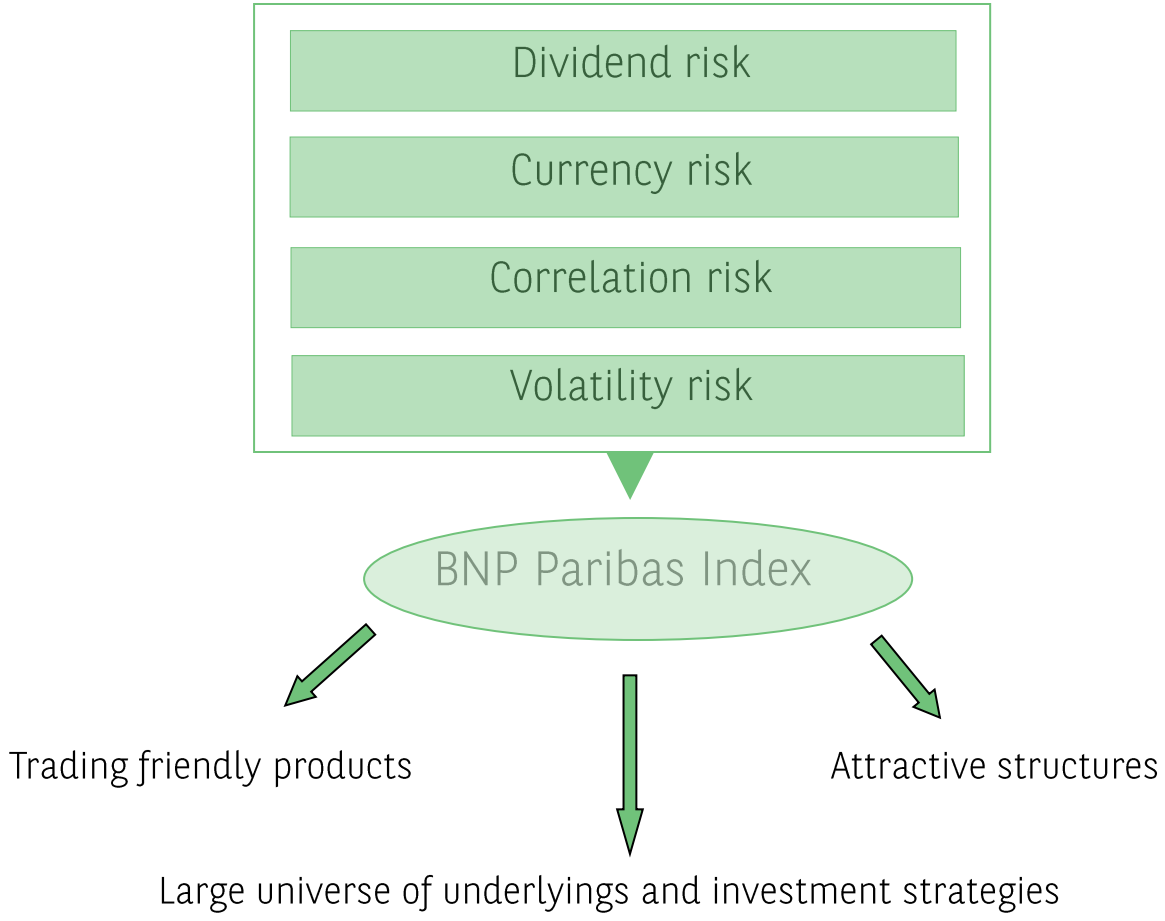
FOREX	FACTOR
BNP Paribas AIR FX ER Index	Volatility
BNP Paribas AIR EUR/USD ER Index	Volatility
BNP Paribas AIR GBP/USD ER Index	Volatility
BNP Paribas AIR JPY/USD ER Index	Volatility
BNP Paribas AIR AUD/USD ER Index	Volatility
BNP Paribas AIR CAD/USD ER Index	Volatility
BNP Paribas Galaxy G10 Excess Return USD Index	Carry
BNP Paribas Galaxy Diversified Excess Return USD Index	Carry
BNP Paribas Galaxy Regional Excess Return USD Index	Carry
BNP Paribas Galaxy World Excess Return USD Index	Carry
BNP Paribas STEER™ G10 USD Index	Value
BNP Paribas FX G10 Time Series Momentum ER Index	Momentum
BNP Paribas MGI Systematic Currency Alpha Index	Multi-Factor

RATES	FACTOR
BNP Paribas AIR T-Note ER Index	Volatility
BNP Paribas AIR Bund ER Index	Volatility
BNP Paribas TIER Bund Index	Volatility
BNP Paribas TIER T-Note Index	Volatility
BNP Paribas Rolling Call US T-Note ER Index	Volatility
BNP Paribas Rolling Call Bund ER Index	Volatility
BNP Paribas Gamma Weighted Long Rates Volatility USD ER Index	Volatility
BNP Paribas Global Government Bond MN Carry Index	Carry
BNP Paribas Global Government Bond MN Relative Value Index	Value
BNP Paribas Global Government Bond MN Momentum Index	Momentum
BNP Paribas Kinetis Money Markets Atlantic ER USD Index	Trend
BNP Paribas Kinetis Bond Futures ER USD Index	Trend

CREDIT	FACTOR
BNP Paribas Credit Compression US ER Index	Carry
BNP Paribas Credit Compression Europe ER Index	Carry
BNP Paribas Credit Compression MN Europe ER Index	Carry
BNP Paribas Credit Compression MN US ER Index	Carry
BNP Paribas Credit Time Series Momentum US ER Index	Trend
BNP Paribas Credit Time Series Momentum Europe ER Index	Trend
BNP Paribas Credit Low Volatility picking Europe ER Index	Defensive
BNP Paribas Credit Low Volatility picking US ER Index	Defensive

OFF-THE-SHELF | SINGLE STRAT | #FACTORS/ARP | INFRA

Index modelling challenges



Different versions are available

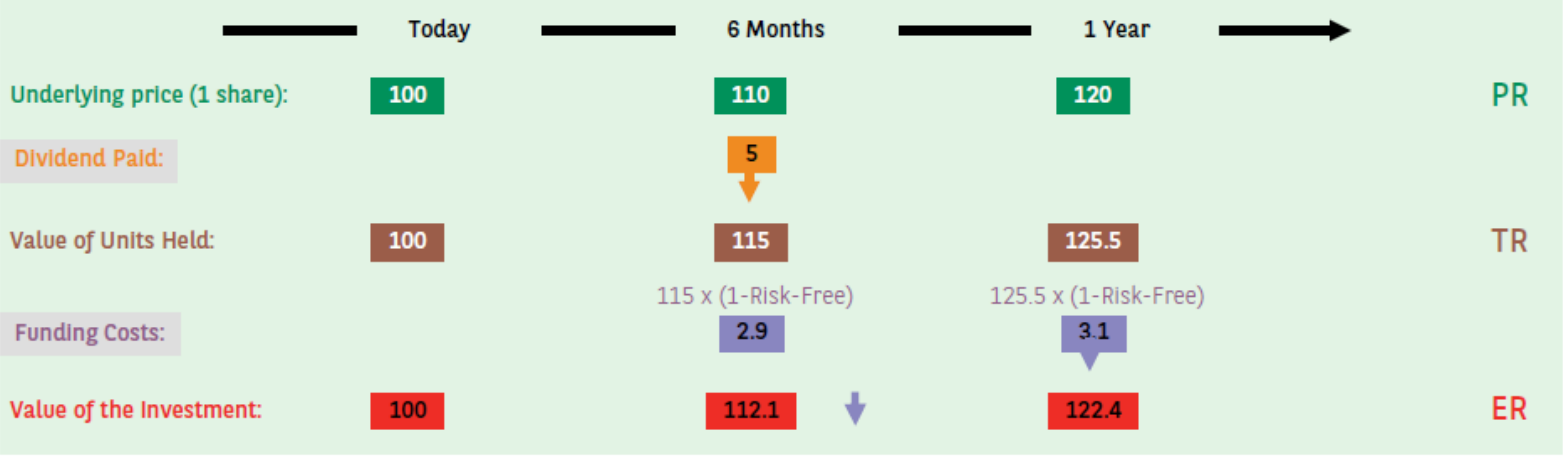
What are the differences?

- The differences between index versions occur when calculating the index level
- All versions of our indices include the gain or loss on the underlying
- Indices can also include one or more of the following in calculating their value:
  - Dividends
  - Funding costs
- Structured products are usually built on Excess Return indices
  - No constraint related to dividends
  - Cost efficient

Breakdown of the differences amongst index versions

Type	Price Return (PR)	Total Return (TR)	Excess Return (ER)
Gain/Loss on Underlying	Yes	Yes	Yes
Reinvestment of Cash Flows	No	Yes	Yes
Integration of funding costs	No	No	Yes

Example of the calculation



*Assuming Risk-free rate 5% annually

Managing dividend risk

As dividends are systematically reinvested, no need to care about their levels

Detailed Mechanism

- T=0 — We buy a stock at EUR 100
- T=6 months — The stock is worth EUR 110
- T=6 months +1 day — The stock pays out a EUR 5 dividend
- T=6 months +1 day — The stock is worth EUR 105

On T=6 months, we reinvest the €5 dividend to buy 5/105e of stock.
Thus we hold **(1+ 5/105e) stock in our portfolio**

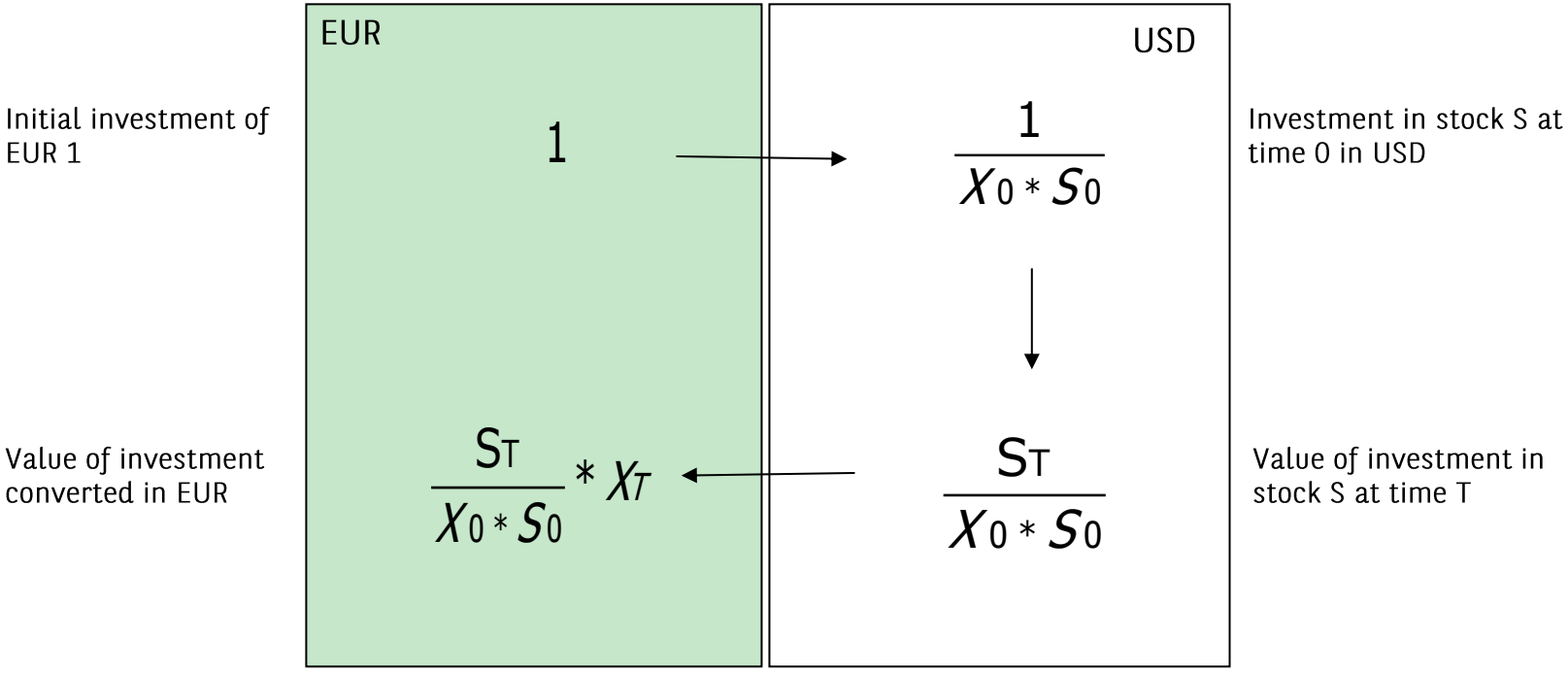
- T=1 year (Maturity) — The stock is worth EUR 120

(1+5/105e) of stock is worth EUR 120, i.e. the performance is (1+5/105) x (120/100)

Managing foreign exchange risk (1/2)

First Step - Investment of 1 EUR in a foreign stock (e.g. USD)

- Exchange rate: X_0



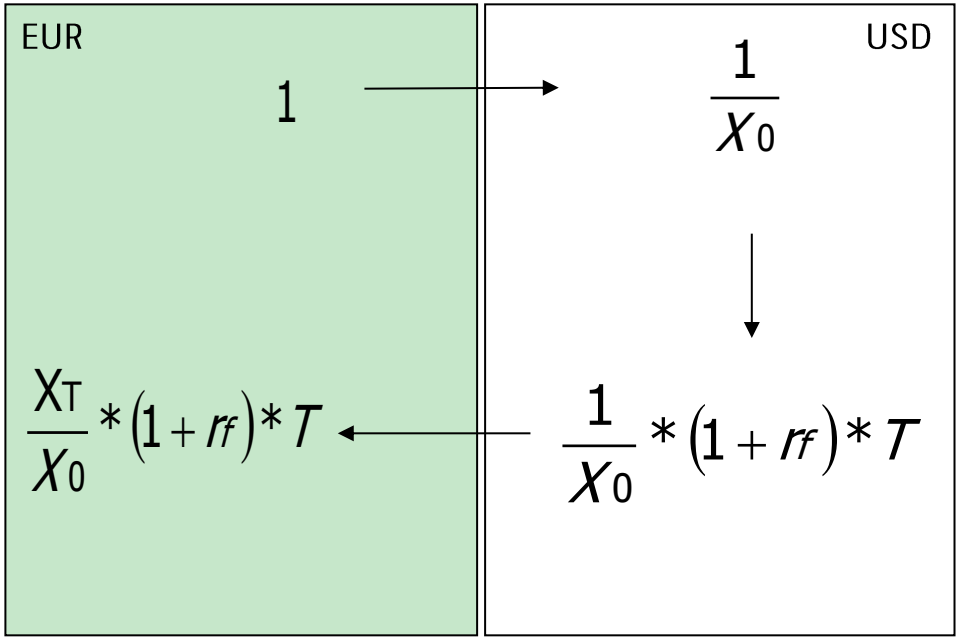
.... Performance is guaranteed but the foreign exchange risk remains ...

Managing foreign exchange risk (2/2)

Second Step - Investment of EUR 1 at the USD risk-free rate

Investment in EUR of the loan amount in USD

Capitalised investment at risk-free rate + converted into EUR



Loan worth 1/X0 USD

Amount to be refunded in USD

Conclusion: the foreign exchange risk is covered

$$\frac{S_T}{X_0 * S_0} * X_T - \frac{X_T}{X_0} * (1 + r_f) * T$$

Managing volatility risk (1/3)

Objective

- Risk control mechanism aims to control the risk of an underlying
- Volatility is used as measure of risk
  - Aims to maintain volatility at or below a target level

Rational

- Historically volatility has risen when markets have fallen
  - Volatility therefore can be used as an indicator of risk
- Reducing volatility reduces the price of structure products on the index
  - Improves the risk return profile of structured products

Mechanism

- The mechanism aims to dynamically adjust exposure to the underlying depending on the volatility
  - Decrease exposure when volatility is high, move to risk-free asset
  - Increase exposure when volatility is low
- Mechanism has the possibility of leverage, and a maximum exposure

$$\text{Target Exposure} = \frac{\text{Target Volatility}}{\text{Realised Volatility}}$$

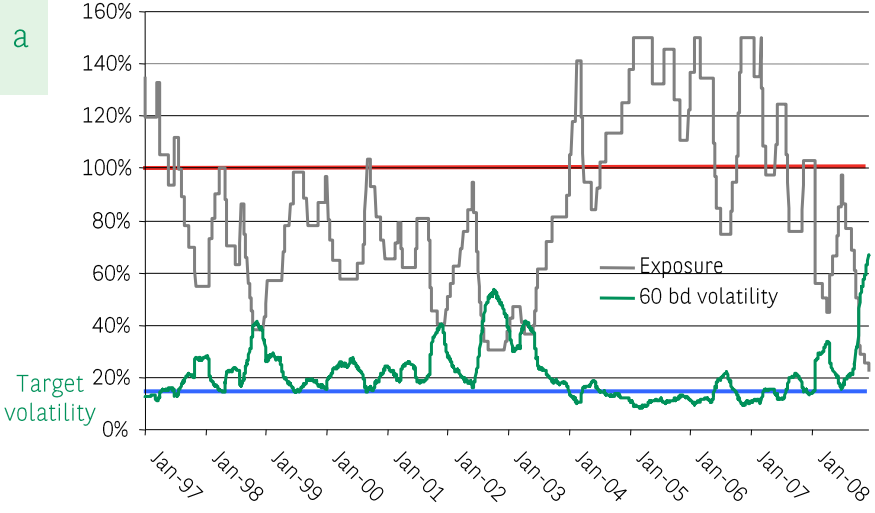
Demonstration of the mechanism

Calculated on: 60 day volatility of the EURO STOXX 50

Target volatility: 15%

Maximum exposure: 150%

Risk-free asset: EONIA



Managing volatility risk (2/3)

Target volatility level: 10%

Maximum exposure to the index: 150%

Scenario 1

Volatility = 5%

Exposure = 150%

(The lower between 150% and 10%/5%)

Scenario 2

Volatility = 10%

Exposure = 100%

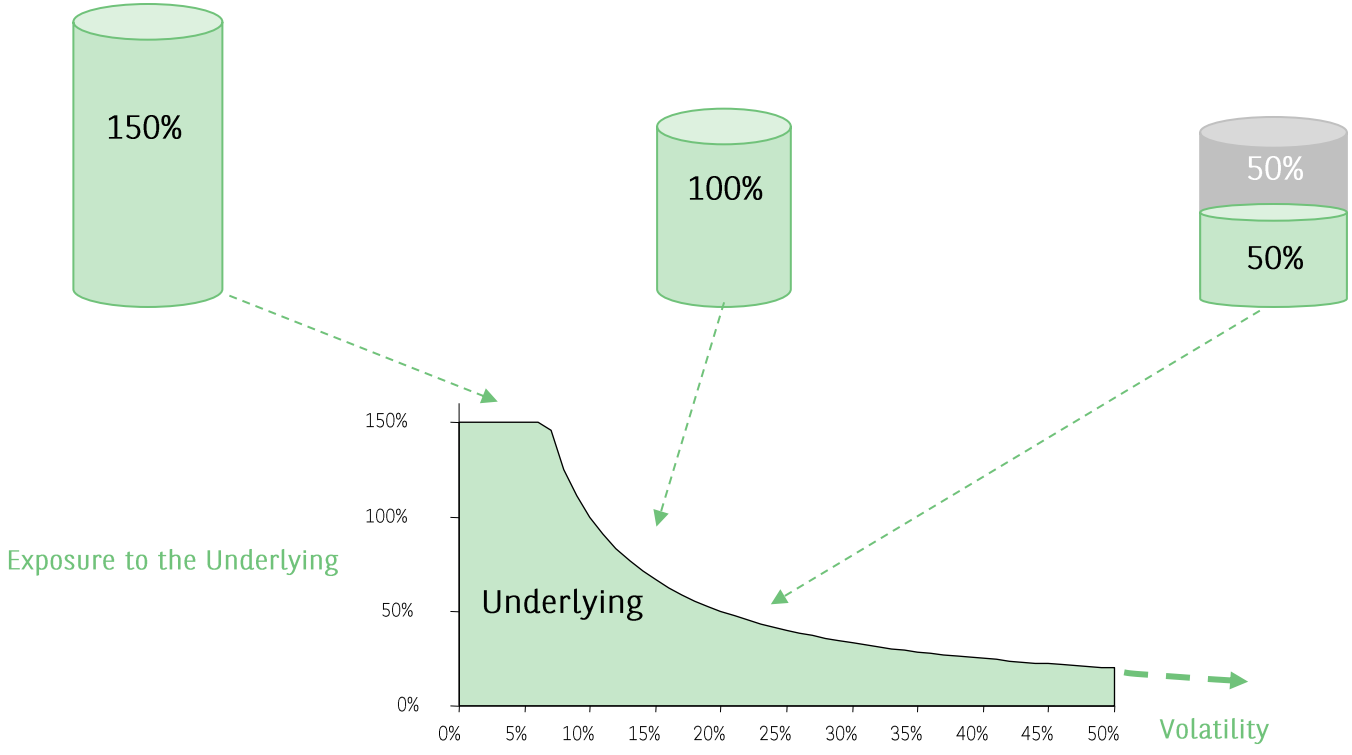
(The lower between 150% and 10%/10%)

Scenario 3

Volatility = 20%

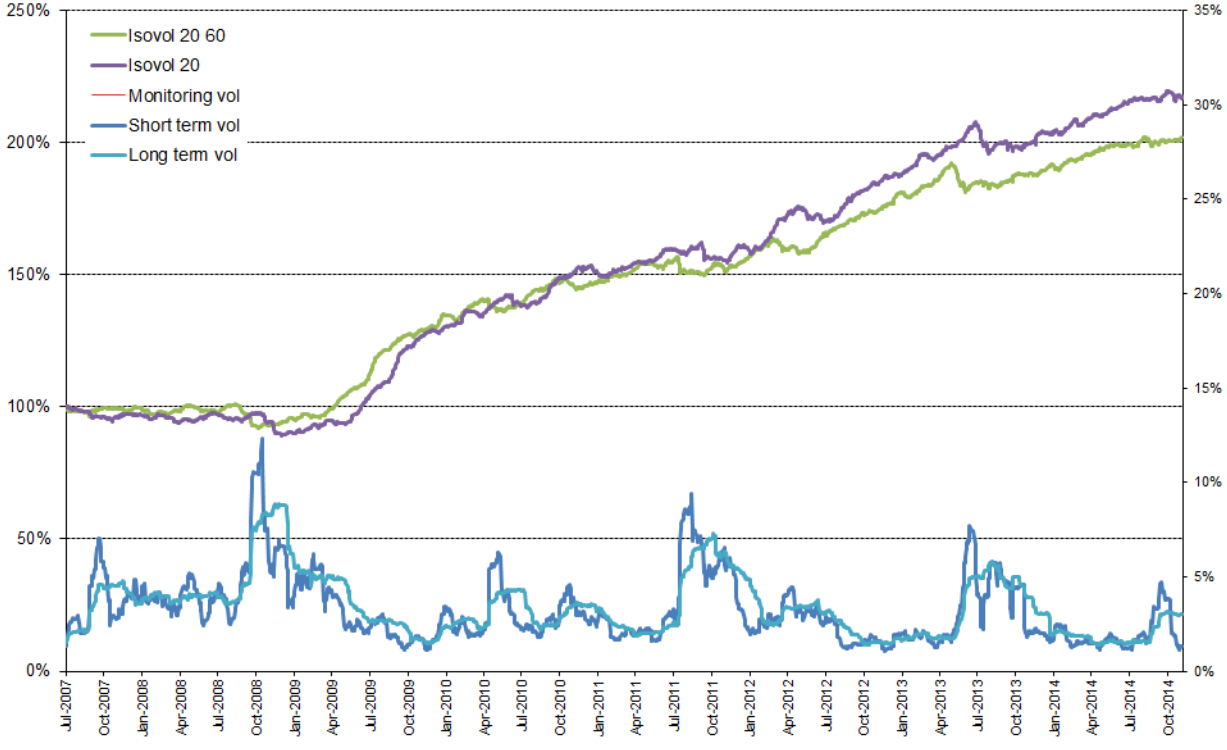
Exposure = 50% (50% in risk-free asset)

(The lower between 150% and 10%/20%)



Managing volatility risk (3/3)

	Comparison			
	Fund	Isovol 20/60	Isovol 20	Isovol 60
Annualized perf	8.72%	9.88%	10.91%	9.75%
Annualized vol	3.84%	3.89%	4.20%	4.16%
Sharpe ratio	**2.27**	**2.54**	**2.60**	**2.34**
	CVar	4.25%	4.64%	4.53%



Heavy publication / maintenance process (1/4)

BNP Paribas Platinium 5 USD ER

ID card
Publication
- Daily snapshot
- Historic

Reshuffle
- Historic
- Checks

BNP Paribas Platinium 5 USD ER

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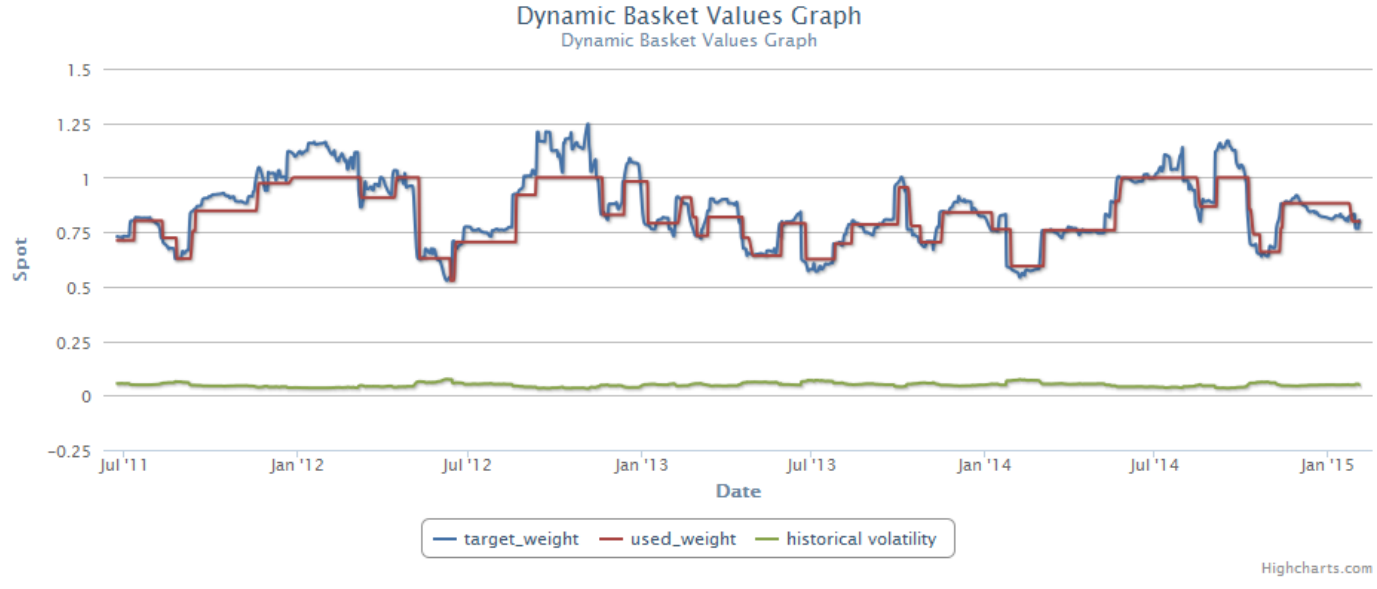
Spots | Spot Records | Checker | User Indicator | BNPIP5UE_IDX | Calendar | Composition Delta Publication | Analysor_FEDOPEN_Capitalized | Configuration

Dynamic Basket Values

Date	spot	risky instrument spot	target_weight	used_weight	coupon_value	historical volatility
02 Feb 2015	150.991925354611	1.74867011066938	0.767199009667344	0.799332089101074	0	0.0521377106799759
03 Feb 2015	150.735939739332	1.74496123327172	0.804072777938216	0.799332089101074	0	0.0497467407148978
04 Feb 2015	151.169651533252	1.75124243807565	0.800543044478268	0.799332089101074	0	0.0499660827433319



Heavy publication / maintenance process (2/4)

KEY AREAS OF FOCUS IN EVALUATING PRODUCTS

- When evaluating Risk Premia strategies and providers, the process and production chain are of the highest importance
- Independent auditors should control and assess all aspects of index processes on a regular basis
- Strong infrastructure should be in place to service after trades

* The evaluation was carried out in accordance with the SOC 1/ISAE 3402 attestation standard, as of 30 June 2018.

01

ROBUSTNESS : AVOID OVERFITTING

- Each strategy goes through a validation process involving functions and independent approval to ensure the quality of the strategy before going live
- All strategies' inputs go through robustness tests performed by an independent research team
- Robustness reports are available on demand

02

EXECUTION & TIMING RISK

- Limit reliance on marketing timing signals
- Look for daily execution and high quality execution platform that allows the provider to mitigate material impact of timing and randomness factor ("a great idea is nothing without proper execution")

03

POST TRADE ANALYSIS & REPORTING

- Explicit and transparent costs with execution commitment
- Detailed performance attribution and regular risk reporting
- Other post-trade service capabilities available upon request

Heavy publication / maintenance process (3/4)

SOC1 ATTESTATION

- **A review of our indices,** in accordance with the **SOC1 / ISAE 3402 attestation standard,** has been periodically carried out **since 2013**
- **The last evaluation was carried out by PwC for the 2018 report (*)**

IOSCO COMPLIANCE

- BNP Paribas issues annually a Statement of Compliance with the "Principles for Financial Benchmarks" (the "Principles") published by the International Organization of Securities Commissions ("IOSCO") on 17 July 2013.
- We have published a **Statement of Compliance with the Principles** (the "IOSCO Statement") **each year since 2014.**
- **In 2018, PwC confirmed BNP Paribas statement is fairly presented.**

BENCHMARK ADMINISTRATOR REGISTRATION

- BNP Paribas SA was registered as a Benchmark Administrator for the purpose of the **EU Benchmarks Regulation (BMR)** by the Autorité des Marchés Financiers (AMF) on 21st May 2019. BNP Paribas is included in the **register of Benchmark administrators** published by the European Securities and Markets Association

Full Marketing Package on BRIO

BNP PARIBAS | brio™

Yann Moysan

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RECENT PUBLICATIONS

Price Efficient — Internal Only

Indice Bloomberg Europe Value Sectors 2023 Decrement 5

l'indice est composé des 30 plus grandes entreprises (par leur capitalisation boursière flottante) opérant dans les secteurs bancaire et automobile de la Zone Euro ainsi que le secteur de l'énergie en Europe.

04 Jan 2024 | Discover Offer

Price Efficient — Internal Only

Indice Bloomberg Europe Value Sectors 2023 Dec 50Pts

04 Jan 2024 | Discover Offer

Equity Payoff — Internal Only

Bond + Vanilla option on China Onshore Index

Bond + Vanilla option on China Onshore Index: CSI500

02 Jan 2024 | Discover Offer

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4104 documents

Title	Summary	Regions	BBG Ticker	Product Type	Themes
Indice Bloomberg Europe Value ...	l'indice est composé des 30 plus grandes ...	EMEA	EVAL23PE Index	Price Efficient	QIS
Indice Bloomberg Europe Value ...		EMEA	EVAL23PT Index	Price Efficient	QIS
Bond + Vanilla option on China O...	Bond + Vanilla option on China Onshore I...	APAC, Hong Kong	CSI500 Index	Equity, Payoff	China Acc
BNP Paribas Vol Momentum 3 In...	BNP Paribas Vol Momentum 3 Index ER ai...	EMEA	BNPXUXM3 Index	Option Strategies	QIS

The two main groups of alternative risk premia

VOLATILITY

CARRY

VALUE

MOMENTUM

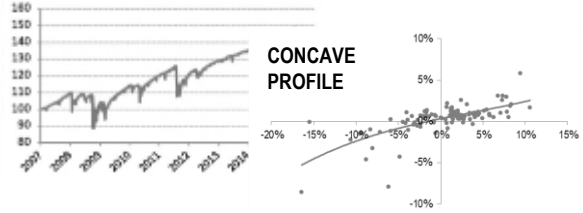
TREND

DEFENSIVE

LIQUIDITY PROVIDING

RISK AVERSION PREMIA

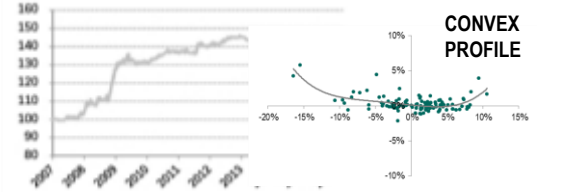
Risk aversion premia are liquidity giving. Implementing these strategies means taking on fundamental risks in the market other investors are adverse to, hence receiving a remuneration. These strategies have typically a ***concave profile***:



BEHAVIOURAL PREMIA

Behavioural premia are liquidity taking. Implementing these strategies means taking advantage of the market anomalies caused by investors' behaviour.

These strategies have typically a ***convex profile***:



For illustrative purposes only.

"<u>SKEWNESS</u> IS INDEED THE MAIN DETERMINANT OF RISK PREMIA"

- Dividends -> crowdedness -> price increase -> downside tail risk
- Downside tail risk -> dearth of option sellers -> increased premium and Sharpe

"NOT ALL EXCESS RETURNS CAN BE CLASSIFIED AS RISK PREMIA"

Lemperiere-Derembe-Nguyen-Seager-Potter-Bouchaud (2014), Risk premia, asymmetric tail risks and excess returns

"<u>VALUE AND MOMENTUM</u> RETURNS CORRELATE MORE STRONGLY ACROSS ASSET CLASSES THAN PASSIVE EXPOSURES TO THE ASSET CLASSES, BUT VALUE AND MOMENTUM ARE NEGATIVELY CORRELATED WITH EACH OTHER"

Asness, Moskowitz, Pedersen (2013), Value and momentum everywhere

"<u>CARRY AND TREND</u> ARE MUTUALLY DIVERSIFYING, ESPECIALLY IN EXTREME STATES. THUS, IT IS INTUITIVELY APPEALING TO COMBINE THEM"

Bhansali, Davis, Dorsten, Rennison (2015), Carry and Trend in Lots of Places "We view this work as highly complementary to that of Asness et al. [2013]"

"THE TERM ALTERNATIVE RISK PREMIA ENCOMPASSES TWO DIFFERENT TYPES OF RISK FACTOR: <u>SKEWNESS RISK PREMIA AND MARKET ANOMALIES</u>"

Hamdan, Pavlowsky, Roncalli, Zheng (2016), A Primer on Alternative Risk Premia

BNPP alternative risk premia Matrix

		EQUITY	COMMODITIES	FOREX	RATES	CREDIT
RISK AVERSION	SHORT VOLATILITY	Implied vs Realized (AIR) Put Selling (TIER) & Strangles, Call Carry, VIX Mean-Reversion	Implied vs Realized Put Selling (TIER Gold) Call Carry	Implied vs Realized (AIR) Put Selling (TIER USD/JPY) Short Strangles	Implied vs Realized (AIR) Short Payers or Puts (TIER) Call Carry	Implied vs Realized (AIR) Short Payers
RISK AVERSION	CARRY	Dividend LS Converse spread High Dividend LS	CDI Carry Dynamic Pre-Roll Alpha	Galaxy (Carry trade)	Global Govies Carry Govies MN Carry FIBE	Credit Beta Indices Compression Compression MN
RISK AVERSION	VALUE	Value LS	Alpha Backwardation CDI Skewness Pairs Value	STEER™	Govies MN Relative Value	
BEHAVIOR	MOMENTUM	Momentum LS	Alpha Momentum		Govies MN Momentum	
BEHAVIOR	TREND	Extremum Reversion Intraday Trend	Time-series Momentum	Kinetis FX G10	Kinetis Money Markets Kinetis Bond Futures Multi-Trend Rates	Momentum
DEFENSIVE	DEFENSIVE	Quality LS Low Volatility LS Vix Term Structure POP Calendar spreads	Daily Dynamic Roll Alpha	Long FX 1Y Vol (Calendar Vol)	Forward Vol Triangle	Low Vol (Quality Income)
DEFENSIVE	HEDGING	RPI & Enhanced Collars Long 1Y Vol VOLA & VOLA Carry Calendar Vol	Delta Replication (VR Series)	DHP ZC Collars	Lower Rates Hedge Inflation Hedge	Long Payers Long Payer Spreads Senior Financial Hedge

BNPP Delta Replication Strategy on Inflation

DELTA REPLICATION :

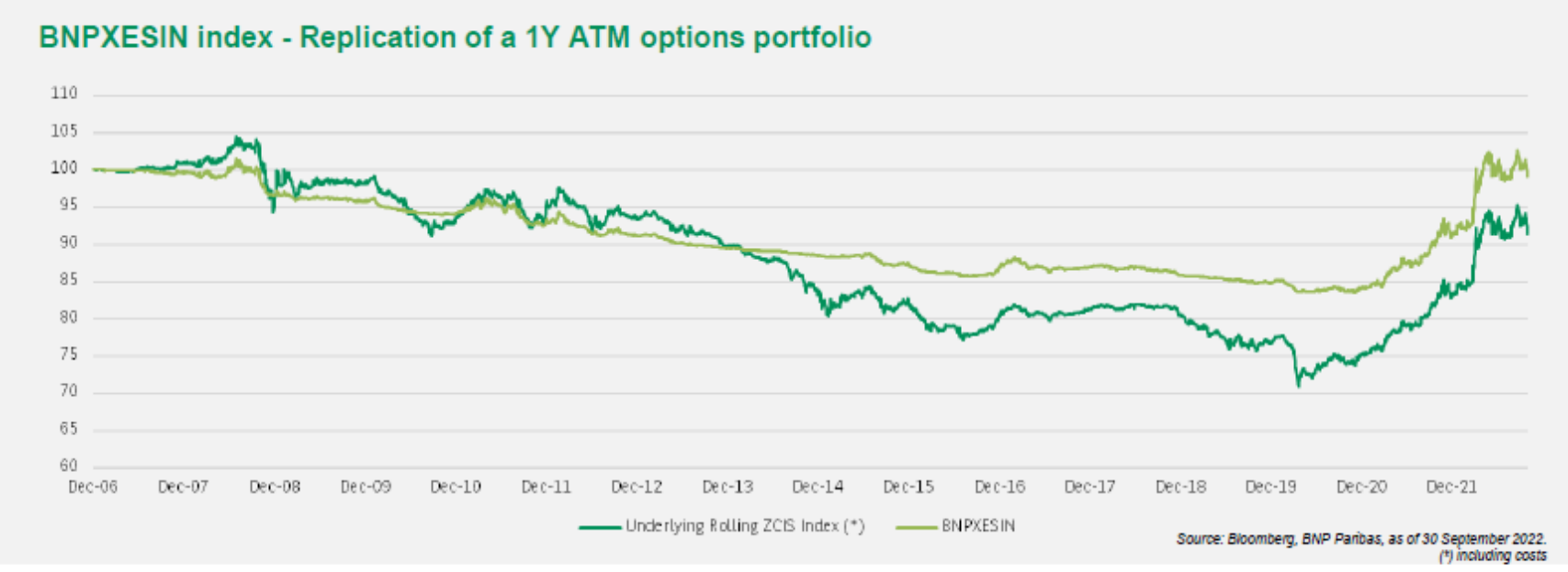
- ALLOWS A NON-LINEAR PAYOFF IN THE ABSENCE OF A LIQUID OPTION MARKET
- EVEN WHEN OPTIONS EXIST, IT IS OFTEN MORE EFFICIENT AS IT SPARES THE VOLATILITY PREMIUM
- HOWEVER, A GAP RISK REMAINS

How it works for inflation breakevens

- The underlying is a 10y Rolling Zero Coupon Inflation Swap Index (after 1 year, the ZCIS which now has a 9y maturity is rolled into a new 10y ZCIS, and the position is split to roll a portion every month)
- A theoretical option portfolio on this index is constituted by adding every day a 1y call on this underlying Index, held to maturity
- The delta of this theoretical portfolio is calculated with the Black-Scholes formula, using a 1y sliding realised volatility as input
- The strategy then takes a position on the underlying index (Rolling ZCIS), that is adjusted every day to match the delta of the theoretical portfolio

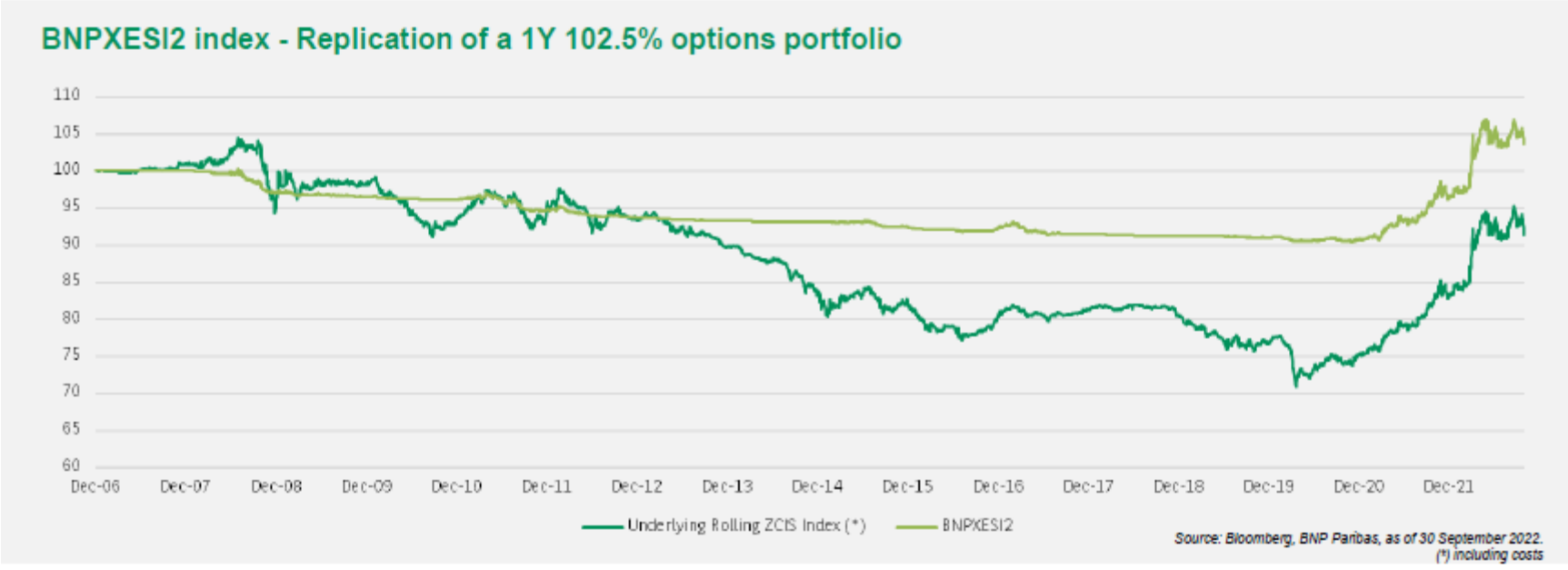
BNPP Delta Replication Strategy on Inflation

BNPP Delta Replication Strategy on Inflation



Period	10Y Breakeven	Underlying Index	BNPXESIN
1 Dec 2006 – 30 Sep 2022	2.18% -> 2.48% (+30bps)	-8.5%	-0.9%
1 Dec 2006 – 30 Dec 2019	2.18% -> 1.20% (-98bps)	-22.3%	-14.8%
30 Dec 2019 – 23 Mar 2020	1.20% -> 0.40% (-80bps)	-8.7%	-1.8%
23 Mar 2020 – 30 Dec 2020	0.40% -> 1.13% (+73bps)	+6.3%	+0.7%
30 Dec 2020 – 28 Apr 2022	1.13% -> 2.94% (+181bps)	+25.2%	+21.4%

BNPP Delta Replication Strategy on Inflation



Period	10Y Breakeven	Underlying Index	BNPXESI2
1 Dec 2006 – 30 Sep 2022	2.18% -> 2.48% (+30bps)	-8.5%	+3.7%
1 Dec 2006 – 30 Dec 2019	2.18% -> 1.20% (-98bps)	-22.3%	-8.9%
30 Dec 2019 – 23 Mar 2020	1.20% -> 0.40% (-80bps)	-8.7%	-0.6%
23 Mar 2020 – 30 Dec 2020	0.40% -> 1.13% (+73bps)	+6.3%	+0.2%
30 Dec 2020 – 28 Apr 2022	1.13% -> 2.94% (+181bps)	+25.2%	+17.8%